2017 EXAMINATIONS

Lancaster University

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.110 Software Development (1 hour & 30 minutes)

- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

Question 1

1.a Variables in the C language are considered statically typed, i.e. types are checked by the compiler at compile time. For each of the following C types describe the nature of the data that can be stored and provide an example of a declaration and initialisation:

i) int
ii) float
iii) char

[6 Marks]

1.b Concisely explain the importance of variable naming. Give two properties of a 'well chosen' variable name. Provide an example of well-chosen variable declarations for:

i) the high score in a game;
ii) the average coursework mark of a student.

[4 Marks]

1.c The C language does not have a 'string' data type. Give a clear, detailed explanation of how strings are handled in C. As part of your answer you should explain how strings are declared and how they are stored. You should include a diagram to illustrate a string in memory as part of your description.

[4 Marks]

1.d You are part of a team developing a 'C library' for an embedded platform, such as the BBC micro-bit. You are part of the 'string team' and have been given the task of writing the 'string length' library function 'strlen' that returns the number of characters in the string passed into it.

Provide a single function implementation of 'strlen' from first principles. You should include the function declaration. As a commercial programmer, naturally you will want to ensure your code follows the recommended house programming style (which just happens to be based on the SCC.110 'C' code style requirements). If you are unable to write detailed code you will still get credit for describing your algorithm.

[6 Marks]

Question 1 continues on the next page…

Question 1 continued.

1.e After a busy day at the office working on 'strlen', you were so galvanised that you thought you'd show your family how to code string functions in C for their pure enjoyment. You decide to write a function that takes a string as a parameter and returns true (1) or false (0) depending on whether the string is palindromic (i.e. reads the same forwards as backwards, examples of palindromic words include “mum”, “dad” and “madam”). Don’t worry about dealing with punctuation - you can assume that the string needs to be exactly palindromic. Provide your solution, making sure you comment your code appropriately.

[5 Marks]

[Total 25 marks]

Question 2

Imagine you have been contracted by the BBC to write a Java program to manage the results of the 2018 Strictly Come Dancing competition. Your program needs to record the total score given to each contestant by a panel of expert judges, and also record the number of telephone votes received from the general public. Based on this information, your program should report the contestants with the highest and lowest overall scores.

2.a Write a Java class to represent a **Contestant**. This class should contain instance variables to hold the following pieces of information. Be careful to choose the best types to hold this information, and ensure your class is properly encapsulated.

- The name of the contestant.
- The score awarded by the expert judges.
- The telephone number members of the public call to vote for this contestant.
- The number of telephone votes received.

[4 Marks]

2.b Write an appropriate constructor for your class. You may assume that no voting or scoring has yet taken place.

[4 Marks]

2.c Write another class called **Strictly**. This should contain a single instance variable – an array of all the contestants in the competition. Show how you could create such a class, and in its constructor, populate the array with instances of your **Contestant** class to hold information on the following contestants:

Name	**Phone**
Nigel Davies	01524 65201
George Boole	11100010111
Alan Turing	76 71 66 84 81
Archimedes PiMan Syracuse	3 14159265

[6 Marks]

Question 2 continues on the next page...

Question 2 continued.

2.d Write a method for your **Contestant** class that allows the judges' score for a contestant to be recorded. Show how your method could be used to enter the following scores:

Name	**Judges' Score**
Nigel Davies	1
George Boole	16
Alan Turing	14
Archimedes PiMan Syracuse	18

[3 Marks]

2.e Write a method for your **Strictly** class called **vote**. This should take a single parameter – a phone number. This method will be automatically invoked every time a telephone vote is received by the BBC. The parameter provides the telephone number that was dialled. In your **vote** method, find the contestant whose phone number matches the one dialled, and increase its vote count by one.

[4 Marks]

2.f The final score given to a contestant is the judges' score multiplied by the number of telephone votes that a contestant received. Show how you can develop your program to calculate the final score for all contestants and print out the contestant with the **highest** score, and the contestant with the **lowest** score. You may create and use further methods as you feel appropriate.

[4 Marks]

[Total 25 marks]

Question 3

3.a

i) There are two common models used for version control software. List both, and explain the difference between them.

[5 marks]

ii) Git uses one of these models. State which model Git uses, and describe how and why Git can apparently implement both of these models.

[4 marks]

3.b The following text is an extract of a diff file in unified format. Explain what each line of the diff is telling us has changed:

```
MIT License
-Copyright (c) 2017 Joe Finney
+Copyright (c) 2017 Joe Finney, Lancaster University
Permission is hereby granted, free of charge, to all
```

[4 marks]

3.c Binary diffs are possible to achieve, but are more complex to handle when a conflict occurs. Explain why this is the case.

[2 marks]

3.d You run your Java code and it crashes, producing the following stack trace for a Null Pointer Exception:

```
Exception in thread "main" java.lang.NullPointerException
    at com.example.Book.getTitle(Book.java:16)
    at com.example.Author.getBookTitles(Author.java:25)
    at com.example.Driver.main(Driver.java:14)
```

i) Where in the code did this exception originate?

[3 marks]

ii) Where was this exception finally caught?

[4 marks]

3.e Describe the difference between black and white box testing; give a notable limitation of black-box tests.

[3 marks]

[Total 25 Marks]

Question 4

4.a Consider the following C program:

```
1  #include <stdio.h>
2
3  int f(int *la, int aL, int *lb, int bL, int *ol)
4  {
5    int oL = 0, ai = 0, bi = 0;
6
7    while (ai <  aL || bi <  bL)
8      if (bi == bL || la[ai] <= lb[bi]) {
9        ol[oL] = la[ai];
10       oL++;
11       ai++;
12     }
13     else {
14       ol[oL] = lb[bi];
15       oL++;
16       bi++;
17     }
18
19   return oL;
20 }
21 /* ARRAYLEN returns the number of elements in an array */
22 #define ARRAYLEN(x) (sizeof(x) / sizeof(int))
23
24 int main()
25 {
26   int a[] = { 1, 1, 2, 4 },
27       b[] = { 2, 3, 5 },
28       c[ARRAYLEN(a) + ARRAYLEN(b)]; /* this is the
29
30   int g = f(a, ARRAYLEN(a), b, ARRAYLEN(b), c);
31
32   for (int i = 0; i <  g; i++)
33     printf("Item %d = %d\n", i + 1, c[i]);
34 }
```

i. Describe as precisely as you can the meaning of the code on the following lines:

- Line 1
- Line 3
- Line 5
- Line 7
- Line 14

[5 Marks]

Question 4 continues on next page…

Question 4 continued.

ii. 'Dry run' the program starting at line 24. Describe the logical flow of execution of the program using the line numbers to help. What is the final state of the output array ('c')?

[6 Marks]

iii. What is the purpose of the program, and what does the program print as a result?

[2 Marks]

4.b Consider the following Java program:

```
1   public class Something
2   {
3       private int[] d;
4       private int[] r;
5
6       public Something(int[] n, int b)
7       {
8           d = n;
9           r = new int[b];
10      }
11
12      public void doSomething()
13      {
14          for (int i=0; i < r.length; i++)
15              r[i] = 0;
16
17          for (int i=0; i < d.length; i++)
18          {
19              int v = d[i];
20
21              if (v < r.length)
22                  r[v]++;
23          }
24
25          for (int i=0; i < r.length; i++)
26          {
27              System.out.print(i + ": ");
28              for (int k=0; k < r[i]; k++)
29                  System.out.print("*");
30              System.out.println("");
31          }
32      }
33
34      public static void main(String[] args)
35      {
36          int[] a = {5,17,0,2,7,3,7,7,1,3,9,13,18,0, 3};
38          Something s = new Something(a,10);
39
40          s.doSomething();
41      }
42  }
```

Question 4 continues on next page…

Question 4 continued.

i) Clearly and concisely explain the purpose of the following lines:

- Line 4
- Line 6
- Line 9
- Line 12
- Line 38

[5 Marks]

ii) Analyse the above program. Describe as precisely and concisely as you can what the program does. Write down what the program would print to the screen.

[7 Marks]

[Total 25 marks]

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